

Kishoreganj University
Department of Computer Science and Engineering
2nd In-course Examination, 2024

Course Title: Artificial Intelligence

Course Code: CSE 3109

Session: 2021-22

Program: B.Sc.

Full Marks: 25

Total Time: 1 Hour

Answer the following questions.

Marks

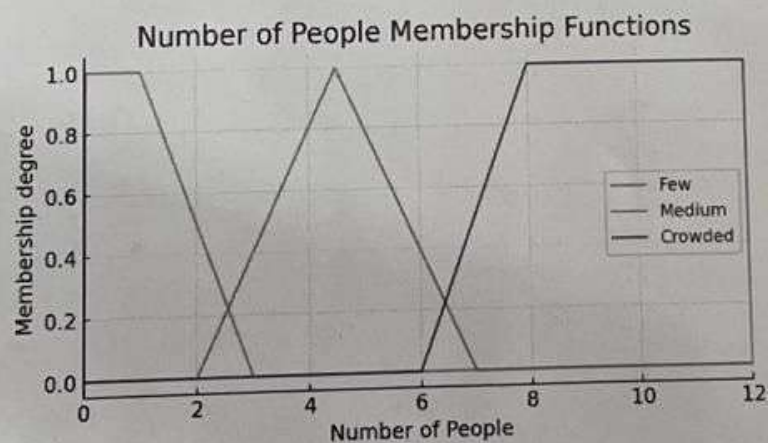
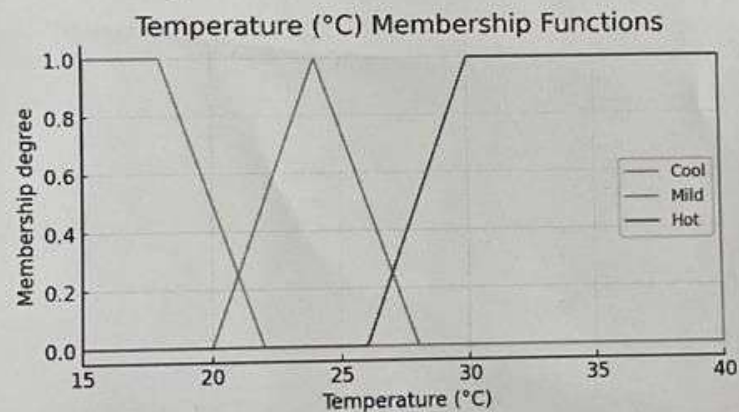
1. In what situations is fuzzy logic more suitable than classical boolean logic? Explain. [4+10]

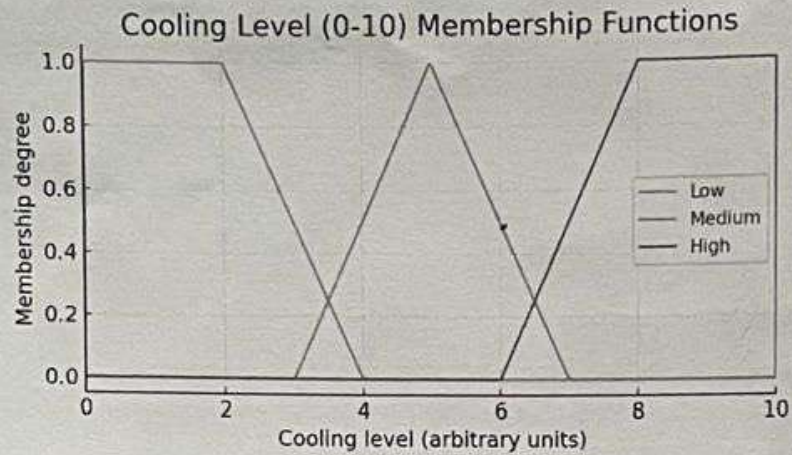
Consider an air conditioner system (AC) that works using the following rules:

- If the room is Hot and Crowded, then High Cooling
- If the room is Mild and Few People, then Low Cooling

How much cooling will the AC apply if room temperature=30°C and number of people=6? (Use **product t-norm** for AND, OR operations and **mean of maxima** defuzzification rule)

Necessary set membership functions are given below.





2. What is knowledge representation? What are the main requirements of a good [3+4] knowledge representation system?

What is predicate logic? Write the following sentence in first order logic (FOL) form:

- i. Mary loves everyone except John.
 - ii. Someone walks and someone talks.
 - iii. Every boy who loves Mary hates every other boy who Mary loves.
 - iv. Anyone who loves everyone loves himself.
3. What is Prolog? Why is it called a declarative language? Describe the features of [4] Prolog.